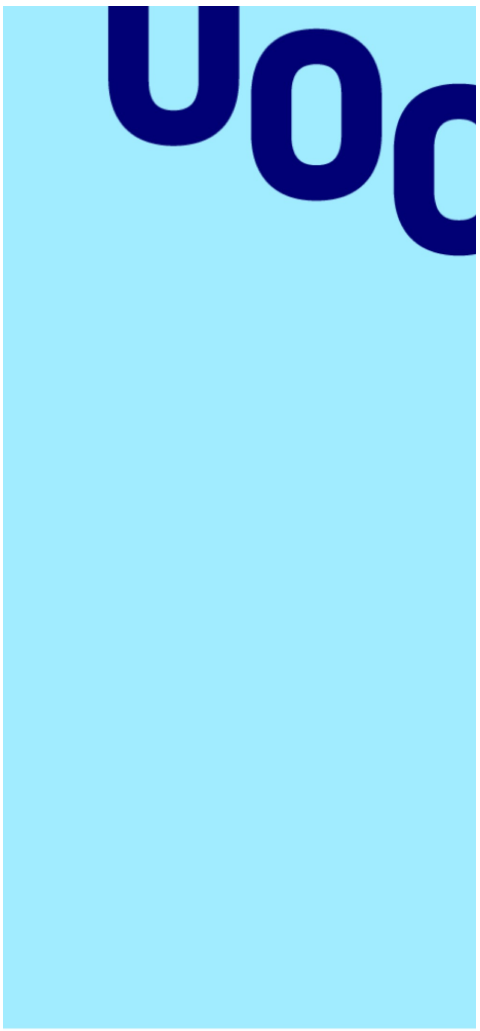
Comparative Evaluation of Satellite Image Segmentation Methods for Coastline Extraction: Spectral Indices vs. Deep Learning.



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Area 4: Pattern Recognition and Computer Vision in Applied Sciences - Geographic Information Systems, satellite images and geopositioning

MSc in Data Science

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March 16, 2026

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Final Work Card

Title of the work:	Comparative Evaluation of Satellite Image Segmentation Methods for Coastline Extraction: Spectral Indices vs. Deep Learning.
Name of the author:	Albert Gallego Jiménez
Name of the tutor:	Carlos Cabezas Rabadán and Joaquín Torres Sospedra
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Keywords:	Earth Observation, Satellite imagery, Coastal Remote Sensing, Semantic Segmentation, Spectral Indices, Water Segmentation

Abstract

A maximum of 250 words, detailing the purpose, context of application, methodology, results and conclusions of the work.

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Chapter 1

Introduction

1.1 Context and justification of work

Coastal regions are among the most dynamic, densely populated, and socio-economically vital areas on Earth. However, these environments are under constant threat from both natural processes and anthropogenic pressures, exacerbated by climate change. As sea levels rise and extreme weather events become more frequent, coastal erosion has emerged as a critical global challenge. Consequently, accurate and continuous monitoring of shoreline position—the instantaneous interface between land and water—is essential for understanding morphological changes, evaluating erosion trends, and planning effective coastal defense strategies.

Historically, tracking shoreline evolution relied on in-situ topographic surveys or aerial photogrammetry. While highly accurate, these methods are often prohibitively expensive, labor-intensive, and limited in spatial and temporal coverage. In recent decades, the advent of Earth Observation (EO) satellites has revolutionized coastal surveying. Missions such as the European Space Agency's (ESA) Sentinel-2 offer free, high-frequency, and multi-decadal imagery spanning the entire globe [**?**]. This wealth of optical satellite data allows researchers and coastal managers to monitor shoreline dynamics at unprecedented scales and at a fraction of the cost.

To extract the exact position of the waterline from a satellite image, traditional methodologies rely heavily on "Spectral Indices". Satellites capture light across various bands of the electromagnetic spectrum, including the near-infrared (NIR) and shortwave infrared (SWIR), invisible to the human eye. Spectral indices are mathematical formulas that combine these specific bands to exploit the differences in how water and land reflect light. For instance, water strongly absorbs infrared light (appearing dark), while land reflects it (appearing bright). By applying these formulas, the original multispectral image is transformed into a single-band intensity map where water bodies are highlighted. Subsequently, a binarization algorithm (like Otsu's method [**?**]) is applied to find an optimal threshold, separating the image into "water" and "non-water" pixels. Finally, to achieve precision beyond the satellite's native resolution, subpixel contouring algorithms (such as marching-squares [**?**]) are used along the boundaries of these binarized masks to plot the continuous shoreline vector. Frameworks based on these principles, such as CoastSat [**?**], SHOREX [**?**], and CASSIE [**?**], have been successfully implemented to derive

large-scale coastal monitoring data.

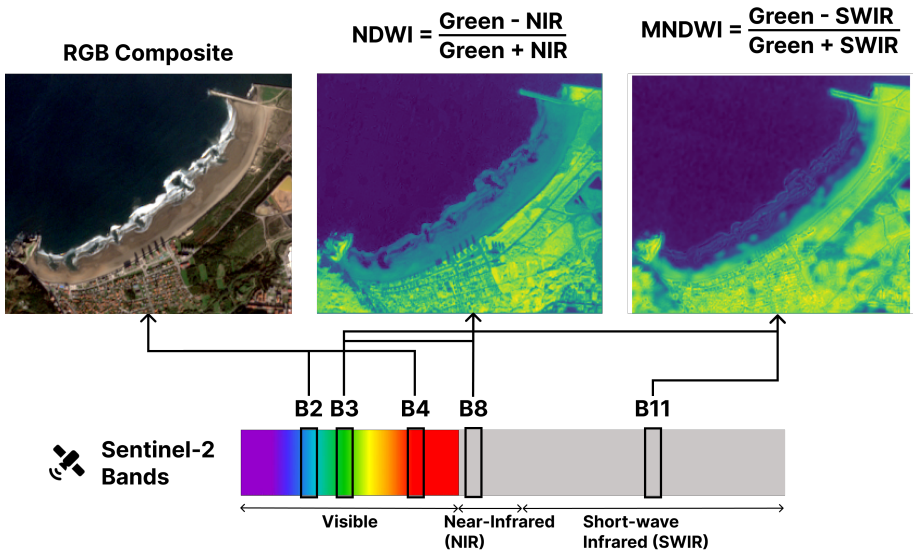


Figure 1.1: Schematic diagram of spectral index calculations using Sentinel-2 bands.

However, this traditional pixel-by-pixel mathematical approach faces severe limitations when applied to highly dynamic and complex environments. On macrotidal coasts, such as those found in the Cantabrian Sea, the large tidal range (up to 4-5 meters) creates extensive intertidal zones. In these areas, the assumption that "water is dark and land is bright" breaks down. Wet, saturated sand during low and ebb tides exhibits spectral signatures remarkably similar to those of shallow water. Additionally, high-energy waves breaking near the shore generate extensive white foam, further confusing the spectral indices. Recent studies (e.g., [**?**]) have demonstrated that these conditions lead to massive landward or seaward biases, as traditional indices fail at the initial segmentation step, incorrectly identifying the wet/dry sand interface or the breaker line as the actual waterline (see Figure **??**).

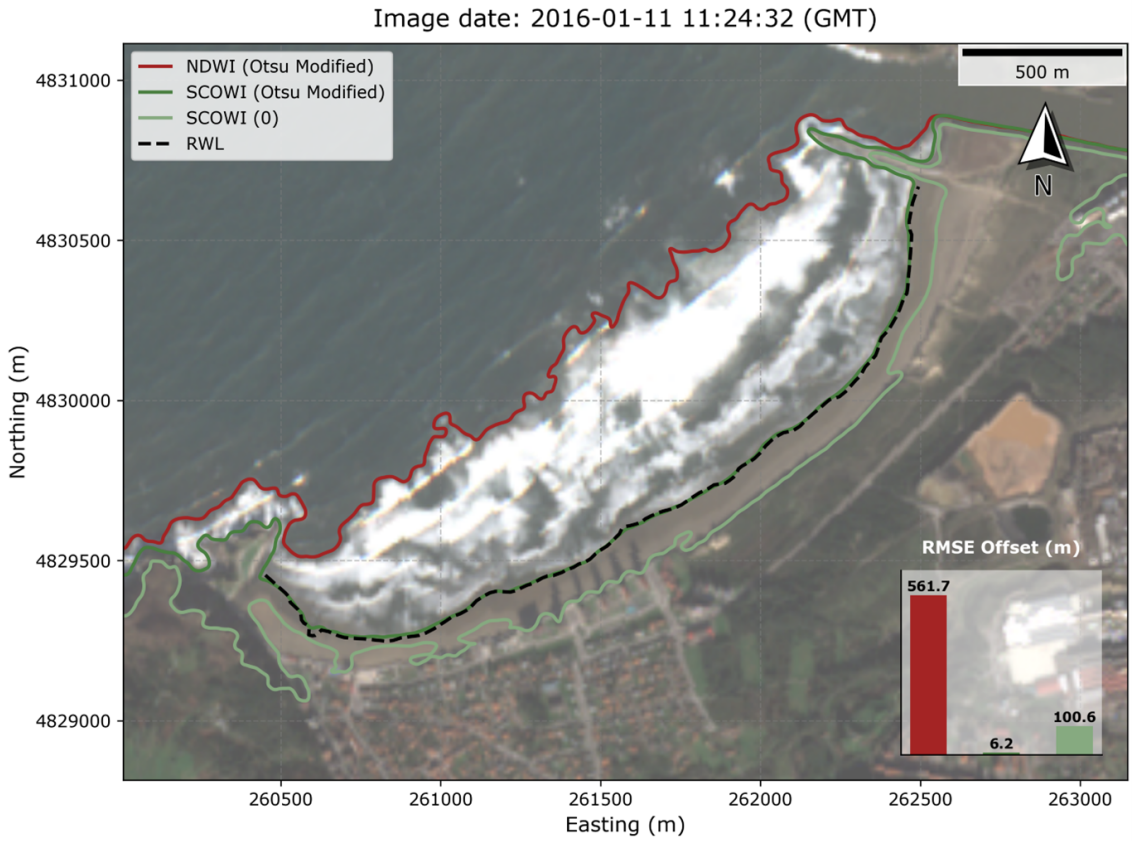


Figure 1.2: Examples of shoreline misidentification in a macrotidal beach (adapted from [**?**])

Consequently, there is a clear and urgent need to move beyond purely spectral formulas and explore methodologies capable of understanding the context of the image. Deep Learning (DL), a subset of Artificial Intelligence (AI), offers a promising solution to bridge this gap. Unlike traditional index thresholds that evaluate the reflectance value of an isolated pixel, advanced DL semantic segmentation architectures can learn complex spatial patterns, textures, and contextual hierarchies. By emulating how the human eye interprets a satellite image—recognizing the texture of a wave or the gradient of wet sand—DL models hold the potential to overcome the spectral confusion inherent in macrotidal environments.

1.2 Work objectives

Primary Objective:

- To perform a comprehensive benchmarking analysis between traditional spectral-based thresholding and Deep Learning-based semantic segmentation for coastline extraction.

Secondary Objectives:

- To generate a high-quality "Ground Truth" dataset of coastal masks using AI-assisted annotation (SAM) [**?**].
- To evaluate the performance degradation of each method under adverse conditions (foam, differences in sand moisture content, and clouds).
- To assess the transferability of trained models between different coastal typologies (Cantabrian vs. Mediterranean).

1.3 Impact on sustainability, ethics-social and diversity

This section should identify the positive and/or negative impacts of the final project on the three dimensions of UOC's cross-disciplinary competence "Ethical and global commitment".

The Cross-cutting Guide on Ethical and Global Competence will help you write these sections.

1.4 Focus and method followed

The methodology is structured into four main phases as illustrated in the workflow diagram (Figure **??**):

1. **Data Acquisition:** Downloading Sentinel-2 Level-2A (atmospherically corrected) products from Google Earth Engine, covering the period 2015–2025. A built-in cloud detection algorithm will be applied as an initial filter to discard obstructed scenes, utilizing the remaining available cloud-free images.
2. **Preprocessing & Labelling:** Calculation of spectral indices (NDWI, MNDWI). Generation of Ground Truth masks using the Segment Anything Model (SAM) through an interactive annotation tool [**?**].
3. **Model Implementation:**
   - *Baseline:* Automatic Otsu thresholding applied to spectral indices.
   - *Deep Learning:* Training semantic segmentation models like U-Net [**?**] or SegFormer [**?**]. Model training will be run locally on an NVIDIA RTX 3060 GPU, with an institutional computing cluster (2 nodes, 2x NVIDIA A30 each) available for scaling if required.
4. **Evaluation:** Comparison using pixel-based metrics (IoU, Dice) and geometric validation (calculating the mean Euclidean distance between the predicted and real coastline).

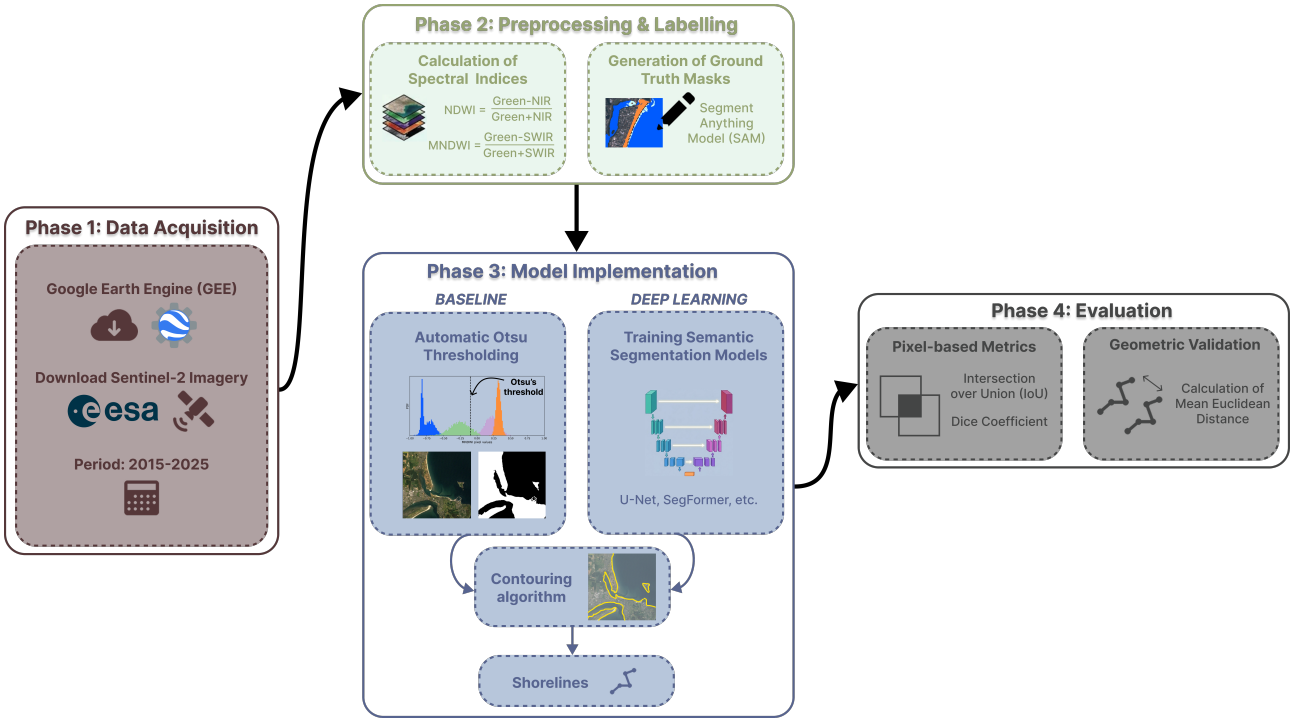


Figure 1.3: Schematic representation of the methodological workflow.

1.5 Work schedule

The project is planned to be developed over a 5-month period (including the thesis defence), structured into the following tasks as shown in the Gantt chart (Figure **??**). Additionally, project management software such as Trello or MS Planner will be used to track and control the tasks at a more specific level:

- **Task 1:** Literature review, selection of study sites (beaches), and satellite images download.
  - Literature review on traditional methods for satellite-derived coastline extraction.
  - Literature review on Deep Learning models applied to coastal monitoring.
  - Satellite images download for Beach 1 (Cantabrian environment).
  - Satellite images download for Beach 2 (Mediterranean environment).
- **Task 2:** Calculation of spectral indices and generation of Ground Truth masks.
  - Preparation of the GitHub repository structure for the project's codebase.
  - Calculation of spectral indices (e.g., NDWI, MNDWI).
  - Create false-color composites to assist in the annotation process.
  - Ground Truth generation through AI-assisted annotation of the dataset.

- **Task 3:** Model implementation: Development of baseline and Deep Learning architectures.
  - Application of automatic thresholding to spectral indices to set baseline models.
  - Selection, development, and training of Deep Learning architectures.
  - Application of contouring algorithms to extract shorelines at sub-pixel precision.
- **Task 4:** Statistical analysis of results, benchmarking across different algorithms, and final evaluation.
  - Pixel-based metrics evaluation (e.g., IoU, Dice Coefficient).
  - Geometric validation by calculating the mean Euclidean distance between predicted and real shorelines.
  - Comparison of performance under adverse conditions (foam, wet sand) and transferability assessment.
- **Task 5:** Drafting report and interim deliveries.
- **Task 6:** Progress meetings with directors.

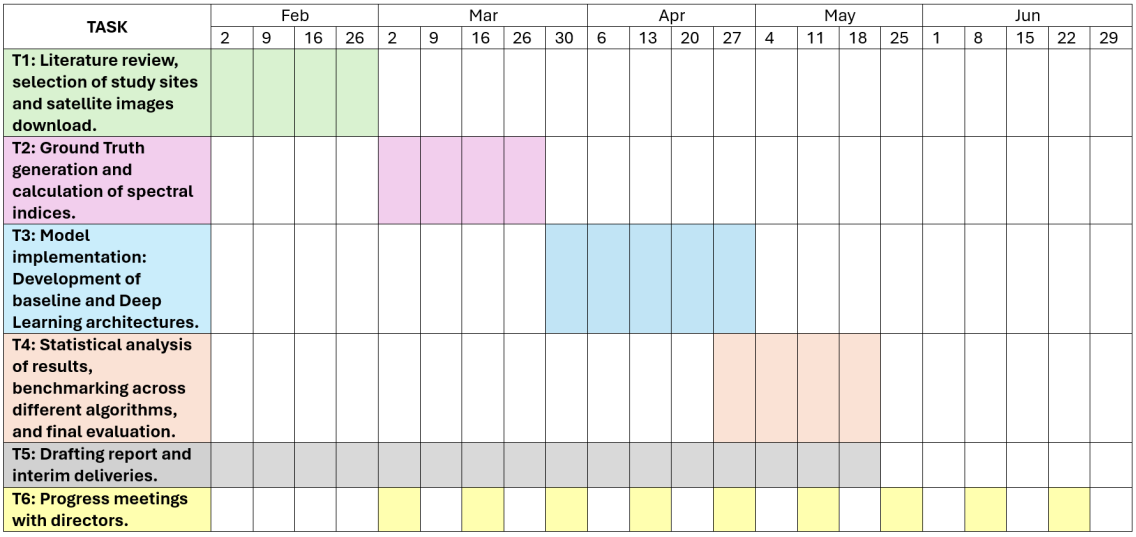


Figure 1.4: Gantt chart of the project timeline.

1.6 Brief summary of products obtained

No need to go into detail: the detailed description will be made in the rest of the chapters.

1.7 Brief description of other memory chapters

Brief explanation of the contents of each chapter and its relationship with the global project.

1.8 Statement on the use of generative artificial intelligence

¡To be filled in¿

Chapter 2

State of Art

2.1 Satellite Remote Sensing for Coastal Monitoring

The ability to extract Satellite-Derived Waterlines (SDWs) has democratized coastal monitoring, enabling both global analyses and detailed local studies. Medium-resolution optical imagery, particularly from the Landsat program (USGS/NASA) and the Sentinel-2 mission (ESA), has become the cornerstone of these efforts. Since 2015, Sentinel-2 has provided unprecedented spatial resolution (up to 10 m per pixel in visible and Near-Infrared bands) with a revisit time of 5 days at the equator, significantly increasing the frequency of observations [**?**].

To leverage this immense volume of data, the scientific community has developed several automated frameworks designed to extract shoreline positions efficiently. Prominent examples include CoastSat [**?**], CASSIE [**?**], and SHOREX [**?**]. These tools typically follow a similar workflow: image acquisition, cloud masking, initial water/land segmentation at the pixel level using spectral indices, and finally, subpixel contouring (e.g., using marching-squares or maximum gradient detection) to define the final shoreline position.

2.2 Traditional Waterline Extraction via Spectral Indices

The critical step in traditional SDW extraction pipelines is the initial segmentation of the image into "water" and "non-water" classes. This is almost universally achieved using Spectral Water Indices. These indices exploit the characteristic of water to absorb most radiation in the Near-Infrared (NIR) and Shortwave-Infrared (SWIR) wavelengths, while land (soil, sand, and vegetation) strongly reflects them.

Historically, the Normalized Difference Water Index (NDWI) and its variant, the Modified Normalized Difference Water Index (MNDWI), have been the standard tools for this task. More recently, indices designed specifically to deal with environmental noise, such as the Automated Water Extraction Index (AWEInsh) and the Subtractive Coastal Water Index (SCoWI) [**?**], have been introduced.

Once the index is calculated to create a greyscale intensity map, a threshold must be applied to binarize the image. While setting a constant threshold (e.g., zero) is common, adaptive statistical methods like Otsu's thresholding are frequently used to find the optimal separation value based on the image's histogram distribution.

2.3 Challenges and Limitations in Macrotidal Environments

While traditional spectral indices combined with automated thresholding (e.g., Otsu) perform highly accurately (often achieving sub-pixel precision of ∼10 m) in microtidal, sheltered, or steeply sloped beaches, their reliability significantly degrades in complex, high-energy, and macrotidal environments. Recent global benchmarking efforts [**?**] have quantified this degradation, showing that horizontal extraction errors more than double (exceeding 20-40 m) on macrotidal beaches compared to their microtidal counterparts. Furthermore, site-specific assessments across the European Atlantic coast have similarly revealed profound limitations in these segmentation techniques during challenging oceanographic conditions [**?**].

The primary driver of error in these settings is the extensive and topographically complex intertidal zone exposed during ebb and low tides. In macrotidal beaches (tidal ranges > 3 m) characterized by intermediate or dissipative morphodynamic states, the retreating tide leaves behind vast plains of saturated sand. Spectrally, the transition from dry sand to wet sand to shallow water is not a sharp boundary but a gradual continuum. Indices such as MNDWI and AWEInsh consistently fail to differentiate between shallow water and saturated soil, systematically misclassifying the wet intertidal zone as "water" [**?**]. This results in the extraction algorithms delineating the upper limit of the wet beach (or even the dune vegetation line) rather than the instantaneous water level, creating massive landward biases that can exceed 50 meters.

Conversely, high-energy wave conditions introduce additional errors. The presence of extensive white foam generated by breaking waves offshore drastically alters the expected low reflectance of the water surface. Indices like NDWI are particularly sensitive to foam, occasionally mistaking the outer breaker zone for land, thus producing severe seaward offsets [**?**].

Ultimately, both specific case studies [**?**] and comprehensive benchmarking exercises [**?**] concur that the primary bottleneck for accurate SDW extraction in complex areas is not the geometric subpixel refinement algorithm, but the initial binary segmentation step (the spectral index determination and thresholding method) itself. Relying solely on the radiometric intensity of isolated pixels is insufficient to navigate the semantic ambiguity between wave foam, wet sand, and the actual waterline. This critical limitation strongly motivates the need to take a step back from classical spectral approaches and explore advanced, context-aware Deep Learning solutions.

2.4 The Shift towards Deep Learning in Earth Observation

To overcome the inherent limitations of pixel-by-pixel spectral thresholding, the remote sensing community is increasingly turning to Deep Learning (DL). DL models, particularly those based on Convolutional Neural Networks (CNNs) and transformer architectures designed for Semantic Segmentation, process images fundamentally differently.

Instead of classifying an isolated pixel based on its specific wavelength reflectance, architectures such as U-Net or SegFormer learn hierarchical spatial features. They consider the context surrounding a pixel—evaluating textures, shapes, and structural patterns. This contextual awareness allows a properly trained DL model to distinguish the texture of wave foam from dry white sand, or recognize the spatial gradient of a saturated beach face without mistaking it for a water body.

Despite the explosive growth of AI in Earth Observation, comprehensive benchmarking studies that strictly quantify the performance disparity between classical spectral thresholding and modern Deep Learning models specifically on challenging macrotidal beaches remain scarce. This project aims to fill that literature gap, presenting a direct, data-driven comparison of these two defining epochs of Remote Sensing methodology.

Chapter 3

Materials and methods

In these sections, it is necessary to describe:

- The most relevant aspects of the design and development of the work.
- The methodology chosen to carry out this development, describing the possible alternatives, the decisions taken, and the criteria used to make these decisions.
- The products obtained.

Chapter structuring may vary depending on the type of work.

If applicable, a section on "Economic evaluation of work" will be included. This section will indicate the expenses associated with the development and maintenance of work, as well as the economic benefits obtained and a final analysis on the viability of the product.

Chapter 4

Results

Detail in this section the results obtained using the methodology described in the previous section.

The figures must be explained and quoted in the text, such as **??**, in which the error is shown depending on the distance, in arbitrary units. All graphs must have the title of the axes.

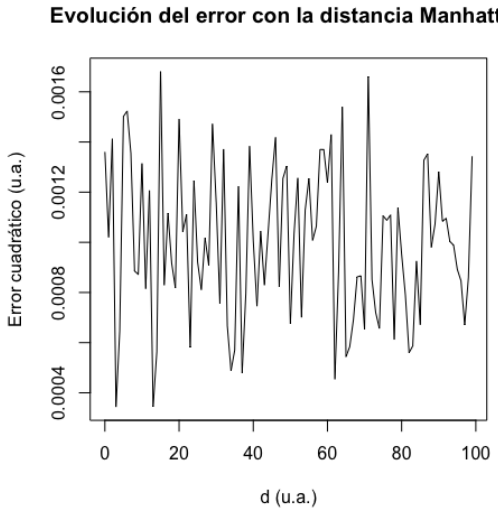


Figure 4.1: Error working distance in arbitrary units.

Chapter 5

Conclusions and future works

This chapter must include:

- A description of the conclusions of the work:
  - Once the results have been obtained, what conclusions are drawn?
  - Are these results as expected? Or have they been amazing? Why?
- A critical reflection on the achievement of the objectives initially set:
  - Have we achieved all the objectives? If the answer is negative, why?
- A critical analysis of the monitoring of planning and methodology throughout the product:
  - Has the schedule been followed?
  - Has the planned methodology been adequate enough?
  - Has it been necessary to introduce changes to ensure the success of the work? Why?
- From the impacts foreseen in **??**, ethical-social, sustainability and diversity, evaluate/mention whether they have been mitigated (if they were negative) or whether they have been achieved (if they were positive).
- If unforeseen impacts have appeared in **??**, evaluate/mention how they have mitigated (if they were negative) or what they have contributed (if they were positive).
- The future lines of work that have not been explored in this work and have remained pending.

Chapter 6

Glossary

Definition of the most relevant terms and acronyms used in the Report.

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List of sections that are too extensive to include in memory and have a self-contained character (e.g. user manuals, installation manuals, etc.)

Depending on the type of work, there may be no need to add an annex.